



NITI Aayog



SECTOR AT A GLANCE 03

Rural: ODF and Waste Management in India

WASH in India

DRAFT

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WASH IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Rural solid and liquid waste management has emerged as an issue of critical importance in India. With rising incomes and consumption, villages are generating increased amounts of both solid and liquid waste. It is estimated that rural India generates between 0.3-0.4 million tonnes of solid waste daily. However, most rural areas have traditionally lacked the infrastructure and systems to effectively manage this waste.

The central government has recognized the management of this waste as a key objective and the SBM-G (2.0) has mandated village-level waste management, behaviour change of communities related to waste management, and also the building of waste management infrastructure in rural India.

This has led to definite progress in terms of capacities with 96% of districts reporting have staff for solid and liquid management, and 88% of the districts having created an end-to-end plastics waste management plan. However, village-level waste treatment infrastructure remains low with only 22% of villages having adequate SLWM infrastructure according to a report by the Standing Committee on Water Resources in 2021-22.

Plastic management is a key problem, especially in the face of increasing FMCG product sales that come packed in single-use plastics from rural India. Most of this is either dumped leading to plastic pileups or burnt causing air and soil pollution. Similarly, source segregation behaviours are poorly adopted.

Going forward, a focus on ensuring sustainable operating models for village-level SLWM, crowding-in private-sector participation, changing citizen behaviours such as segregation, and developing the sanitation O&M ecosystem to ensure proper upkeep of pits and other treatment facilities will be critical for successful waste and grey water management.

This note presents key trends and statistics from within rural solid and liquid waste management, identifies opportunities and challenges, and concludes with some ideas for the way forward.

Figure 1: Key ODF + Rural Waste Management statistics

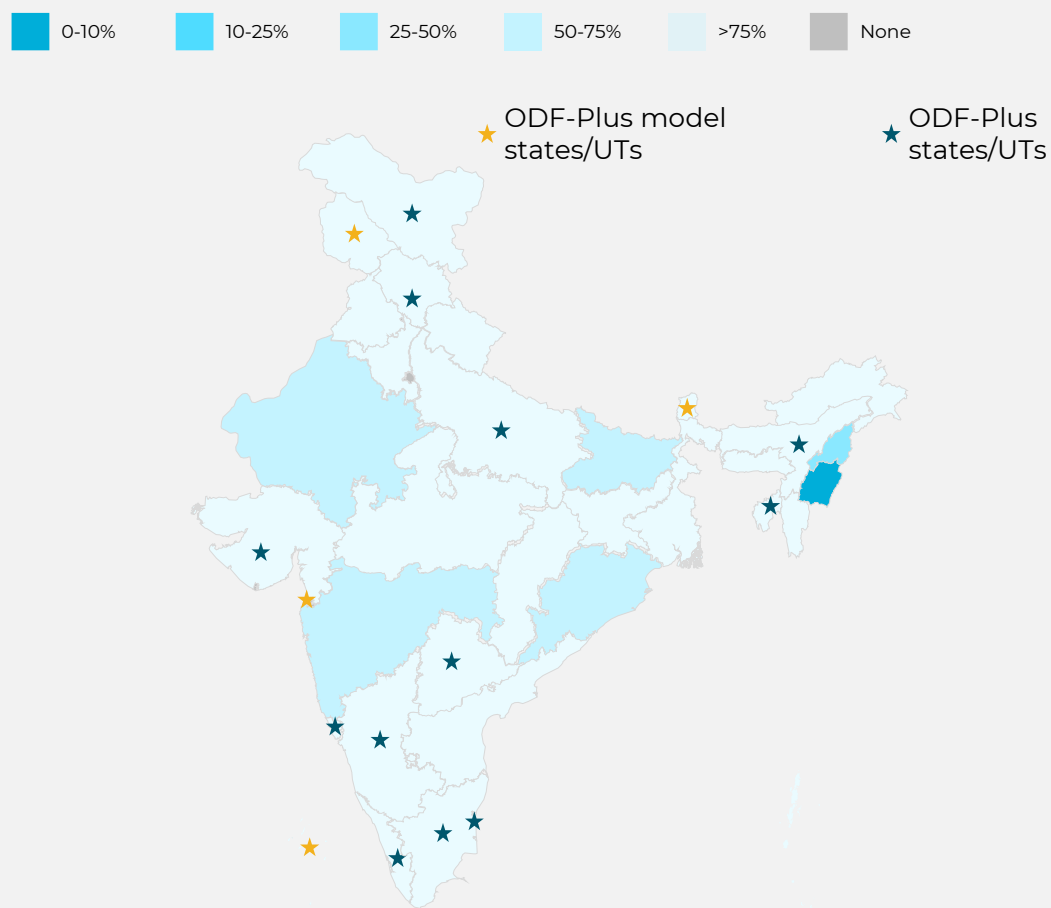
5.08 lakh+ (~87%)
Villages declared ODF Plus*

96,000+
ODF + Model Villages

Source: Ministry of Jal Shakti, SBM-G Dashboard, accessed December 2023

Note: * An ODF Plus village is one which has sustained its Open Defecation Free (ODF) status along with implementing either solid or liquid waste management systems.

Figure 2: Share of villages with ODF-Plus status by states and UTs (2023)



Source: Ministry of Jal Shakti, SBM-G Dashboard, accessed December 2023

Note: * An ODF Plus village is one which has sustained its Open Defecation Free (ODF) status along with implementing either solid or liquid waste management systems. An ODF Plus Model village sustains its ODF status, has both solid and liquid waste management arrangements, observes visual cleanliness, and displays ODF Plus information, education, and communication messages. States with 100% villages achieving ODF-Plus Model or ODF Plus status are shown in the figure above.

Key trends

This section dives deeper into India's rural waste management, outlining statistics and trends on:

1. Solid Waste Management
2. Liquid Waste Management

Solid Waste Management

Solid Waste Management (SWM) activities include collecting and segregating biodegradable and non-biodegradable waste. The management and disposal of bio-degradable waste is done through household and community-level compost pits and bio-gas plants under Galvanizing Organic Bio-Agro Resources Dhan (GOBARDHAN), while management of non-biodegradable waste is undertaken through storage units at the village level and material recovery centres at the block level.

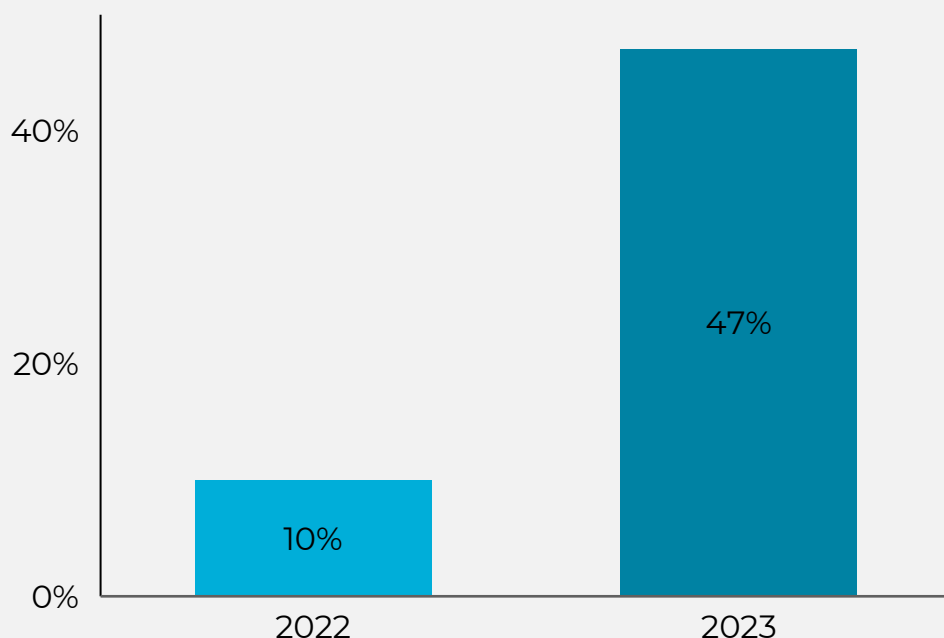
India has made promising strides in SWM in rural areas:

According to the Swachh Survekshan Grameen report, approximately 96% of districts in India have dedicated staff for Solid and Liquid Waste Management (SLWM). Moreover, 88.2% of districts have developed an integrated plan for end-to-end management of plastic.¹

Annual progress on developing waste management infrastructure has been slower compared to targets:

The Standing Committee on Water Resources noted that in 2021-2022, only 22% of villages (46,347 out of 2,07,945) were covered by SWM infrastructure, falling short of the targets set under the SBM-G.²

Figure 3: Share of target villages with arrangements for solid waste management (2022-23)



Source: Ministry of Jal Shakti, SBM-G Dashboard, accessed December 2023

Liquid Waste Management

¹ Department of Drinking Water and Sanitation, Swachh Survekshan Grameen-2022 (SSG 2022), 2022

² Lok Sabha Secretariat, Standing Committee on Water Resources, 2023

Liquid Waste Management or LWM policies and measures in rural India focus on faecal sludge and greywater management. Most rural Indian sanitation is decentralised, and toilet infrastructure and wastewater treatment efforts prioritise low-cost solutions that can be established with low-capacity thresholds.

Prevalence of decentralised treatment options in rural India:

Since all of the sanitation infrastructure in rural India is decentralised (72% served through twin toilets, 10% single toilets, and 18% through septic toilets), treatment options have included deep row entrenchment (trenches), unplanted drying beds, and planted drying beds that reduce the reliance on centralised processing plants.³

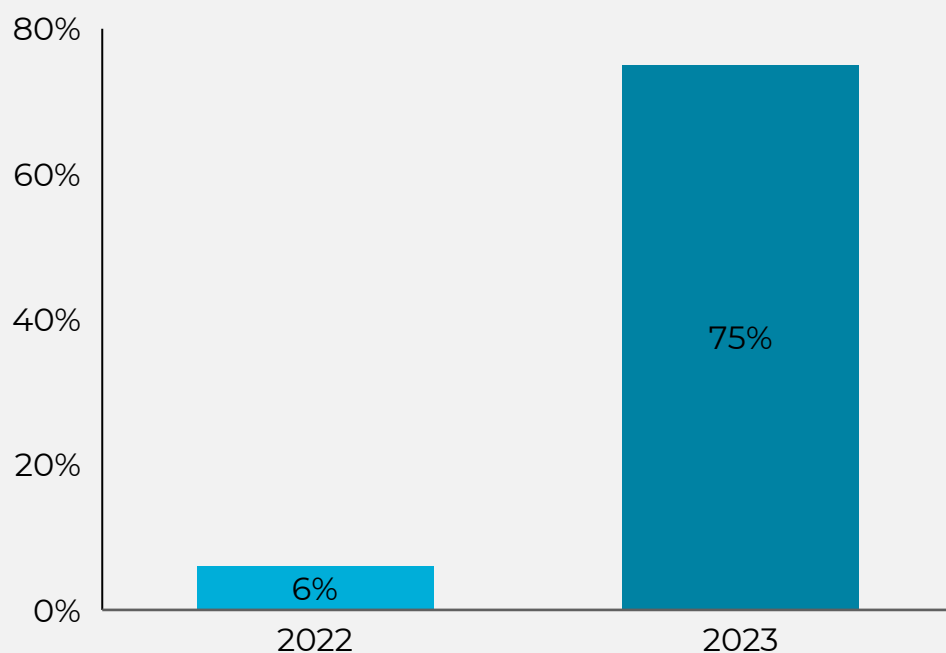
Annual progress for the development of waste management infrastructure has been slower compared to targets:

~12% of target villages (21,734 out of 1,82,517) were covered under LWM infrastructure in 2021-22.⁴

Faecal Sludge Treatment Plants (FSTPs) offer sustainable faecal sludge management:

FSTPs have been set up to serve clusters of gram panchayats, with low per capita capex of ~INR 100-200 and low per capita OPEX of ~INR 10-30, serving as a feasible and sustainable long-term solution to faecal sludge management.⁵

Figure 4: Share of villages with arrangements for liquid waste management (2022-23)



Source: Ministry of Jal Shakti, SBM-G Dashboard, accessed December 2023

³ Ministry of Jal Shakti, Faecal sludge management in Rural areas, 2020

⁴ Lok Sabha Secretariat, Standing Committee on Water Resources, 2023

⁵ WaterAid, *Rural FSM – Journey thus far*, 2020

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective rural waste management practices. These revolve primarily around improved health outcomes, economic benefits, and water security.



Improved health outcomes:

In India, rural areas generate approximately 0.3 to 0.4 million metric tons of solid waste (organic/recyclable) daily.⁶ Improper management of waste poses risks to public health.⁷ By implementing effective waste management practices, significant reduction can be achieved in the occurrence of diseases, leading to improved public health.



Improved economic benefits:

Through efficient SLWM, financial benefits can be achieved by reusing or recycling products that would have been discarded as waste. A report led by the Ellen MacArthur Foundation with the support of ClimateWorks and UNCTAD notes that a circular economy trajectory could bring India annual benefits of USD 624 billion in 2050.⁸



Improved water security:

Sustained wastewater management programs will help ensure a continued and clean supply of potable water to millions. Wastewater management programs in rural areas aimed at treating and managing wastewater, including the reuse of sewage and water used for non-potable purposes, can help prevent pollution of water sources while supporting the sustainability of water tables and sources. By implementing such programs, communities can secure a reliable and clean water supply and enhance water security while benefitting both people and the environment.

Challenges

This section highlights some of the major challenges and constraints facing rural waste management in India. Efforts to resolve such challenges will ensure reduced pollution, prevention of diseases, and resource recovery.



Low adherence to source segregation practices:

Over half of all households nationwide (53%) reported having disposed of the waste outside a common system while approximately 14% reported that the garbage was disposed of indiscriminately.⁹ As per the Ministry of Drinking Water and Sanitation, though solid waste generated in rural areas is predominantly organic and biodegradable, the lack of segregation at source is a growing concern.¹⁰ The Ministry also points to inadequate functional waste collection and disposal systems at the Panchayat level in India.¹¹



Wastewater transportation issues:

Latrines and septic pits are not always accessible to faecal sludge operators and trucks. They are also located too far from a Sewage Treatment Plant or FSTP, leading to high cost of transportation. The roads can also be too narrow and poorly constructed for emptying vehicles.¹²

⁶ National Institute of Rural Development & Panchayati Raj, *Solid Waste Management in Rural Areas*, 2016

⁷ MoUD, *Municipal solid waste management Manual*, 2016

⁸ EllenMacArthur Foundation, *Circular Economy in India: Rethinking Growth For Long-Term Prosperity*, 2016

⁹ Ministry of Jal Shakti, NARSS, 2019-20

¹⁰ National Institute of Rural Development and Panchayati Raj, *Solid Waste Management in Rural Areas*, 2016

¹¹ National Institute of Rural Development and Panchayati Raj, *Solid Waste Management in Rural Areas*, 2016

¹² National Institute of Urban Affairs, *Faecal Sludge and Septage Management — An Advanced Training Module*, 2018

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee rural waste management.



Ministry of Jal Shakti¹³

The Ministry is responsible for the implementation of the Swachh Bharat Mission-Grameen 2.0. It also houses the Department of Drinking Water and Sanitation (DDWS), which oversees rural waste management initiatives in India, more specifically sustaining the sanitation behaviour change that has been achieved under the Swachh Bharat Mission Grameen and increasing access to solid and liquid waste management.



Ministry of Panchayati Raj¹⁴

The Ministry is responsible for all matters related to the Panchayati Raj and Panchayati Raj Institutions in rural areas. It collaborates with the Ministry of Jal Shakti to implement rural sanitation initiatives.

Major policies and programs

This section highlights existing policies related to rural waste management and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI for States platform.



Swachh Bharat Mission-Grameen 2.0¹⁵

The Mission supports the achievement of ODF+ goals by provisioning separate funding for targeted IEC and behaviour change campaigns around waste management practices at household levels. It also emphasises the creation of community assets for solid, liquid, and plastic waste and FSM and ensuring their operations & maintenance (O&M) for sustainable use in the future.

¹³ Ministry of Jal Shakti, Department of Drinking Water and Sanitation, Accessed December 2023

¹⁴ Ministry of Panchayati Raj, Joint Advisory on Swachh Bharat Mission-Grameen (SBM-G) Phase II, Accessed 2023

¹⁵ Ministry of Jal Shakti, [Swachh Bharat Mission-Grameen 2.0](#), 2022

Way forward

India has reached an important inflexion point in its rural waste management journey. Thanks to programs such as SBM-G, significant progress has been made over the past decade to construct toilets and achieve ODF status across Indian villages. Significant progress has also been made in terms of staff capacity to handle rural SLWM.

Going forward, progress for solid and liquid waste management in rural India can be catalyzed through focused efforts on four key areas:

 Ensuring village coverage for SLWM infrastructure	• Ensure access to SLWM infrastructure across villages using decentralized or clustered systems like FSTPs. This can be done by developing sustainable financing models for O&M of decentralized systems through innovative blends of user fees, CSR and philanthropic grants. Additionally, government incentives can help ensure sufficient capital for village infrastructure and secure sustainable funding streams for ongoing O&M expenses. • Policymakers may also consider enhancing grey-water management by promoting the use of simple, low-cost treatment systems that can be easily maintained by the local community. This may include educational programs to raise awareness about the benefits of greywater reuse in agriculture and household chores, thereby reducing the reliance on freshwater sources. Providing subsidies or financial assistance for the installation of these systems can make them more accessible to rural households.
 Enhancing data-driven decision-making	• Create detailed maps and analyses of waste generation patterns, existing infrastructure, and gaps at the village and Gram Panchayat level • Gather data on waste composition and current treatment methods to inform stakeholders and develop better solutions
 Fostering behaviour change and community engagement	• Enhance behaviour change programming to promote practices like waste segregation and safe disposal • Implement sustained Information, Education, and Communication (IEC) campaigns highlighting health risks to elicit greater community participation
 Enabling corporate partnerships for plastics waste management	• Form partnerships with FMCG companies under Extended Producer Responsibility (EPR) frameworks to facilitate collection and environmentally sound disposal of plastic waste and support transportation costs and recycling infrastructure development through even small percentage points of the end product cost, bridging existing resource gaps

Recommended readings

Ministry of Jal Shakti. (2023). Swachh Bharat Mission Grameen Phase II.

<https://static.pib.gov.in/WriteReadData/specificdocs/documents/2023/jun/doc2023616213901.pdf>

Ministry of Jal Shakti. (2022). Swachh Survekshan Grameen 2022

<https://swachhbharatmission.gov.in/sbmcms/writereaddata/Portal/Images/pdf/SSG-2022-report.pdf>

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